STREAMLINING DOCUMENT VALIDATION FOR NJ BUSINESSES

State of New Jersey | NJ Innovation Authority - Resident Experience

Kapila Mane, Product Management Fellow | *New Jersey Institute of Technology, Human-Computer Interaction (HCI)*

Keywords:

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Summary:

To reduce application processing times and accelerate grant + funding approval for New Jersey's business owners, Kapila worked with the New Jersey Innovation Authority as a product manager to design and implement an AI-powered document validation tool for the NJ Economic Development Authority (NJEDA). The tool enables applicants to identify and correct errors—such as illegibility, missing information, or incorrect file formats—the moment a document is uploaded. Kapila's work included **interviewing users**, **evaluating integration approaches**, **defining product requirements**, and **establishing metrics** to track documents flagged and corrected before entering the review pipeline. This solution aims to save time for reviewers and applicants alike, while improving the applicant experience for EDA's application process.

Streamlining Document Validation for New Jersey's Businesses

New Jersey Innovation Authority

Resident Experience (ResX)

Tristan Vanech — Product Manager

coding it forward >

NJ / Innovation Authority

KAPILA MANE

INTRODUCTION

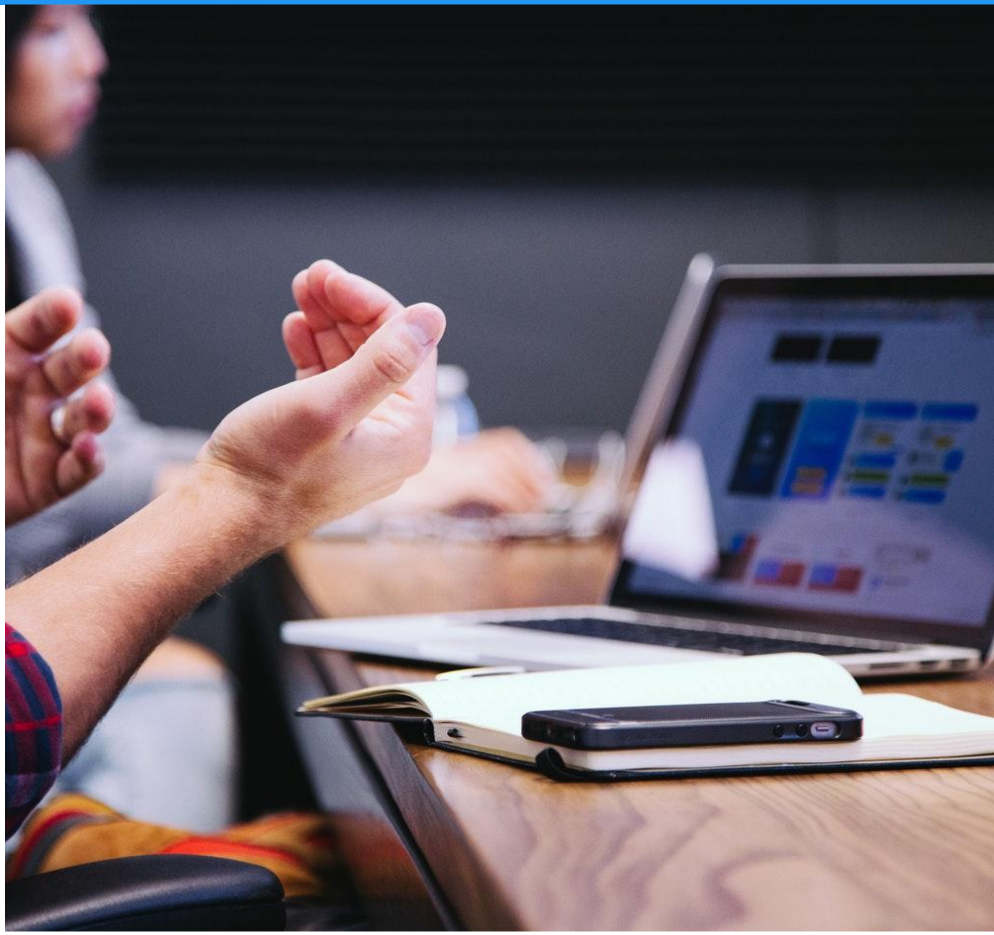
Agency and problem context

WHAT IS THE NJIA?

- The **New Jersey Innovation Authority (NJIA)** is a state entity aimed at efficiently **optimizing and modernizing government** functions by partnering with state agencies
    - Codified into law in Jan 2026
- Divided into two initiatives: Resident (ResX) and Business (BizX)

WHAT IS THE NJEDA?

- The **New Jersey Economic Development Authority (NJEDA)** is responsible for the state's long-term **economic growth**
    - Offers grants and tax relief programs for businesses throughout the year
- Reached out to NJIA



THE CHALLENGE

- EDA's opportunities are valuable for business owners across the state and thus receive thousands of applicants
- Time from initial submission to EDA approval = **cure cycle**
  - Applicants may submit incorrect documents, high rate of **preventable** errors
  - Back-and-forth email chain between applicant and EDA reviewer
  - A cure cycle can take anywhere from a **couple of months to a year!**

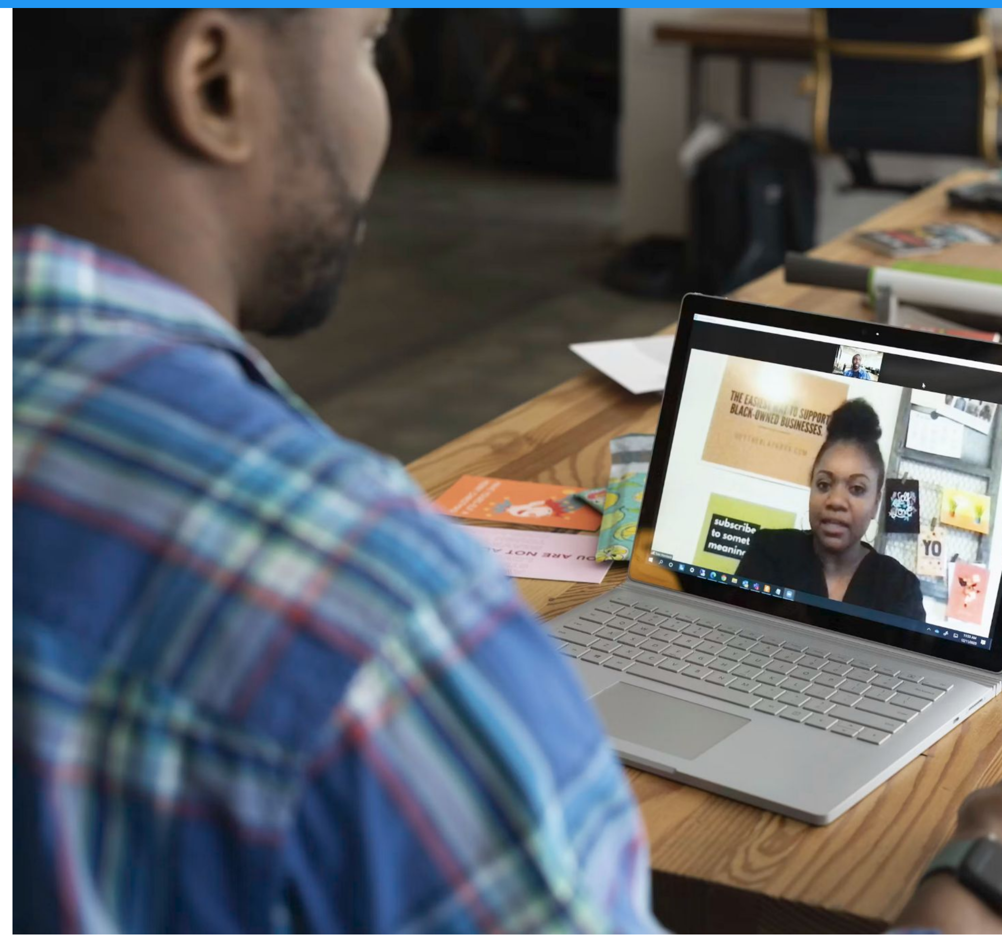
How might we help **applicants** submit error-free documents the first time, so they receive funding faster?

DISCOVERY

Understanding the journey of an application from both sides

COLLECTING DATA

- Designer and I interviewed EDA staff and previous + first-time applicants, qualitative coding in **Dovetail**
- Engineer analyzed all email chains to identify common keywords and recurring issues
  - Used a local AI model **(Google Gemma)** on a 5-10% sample to understand applicant sentiment



INTERVIEW FINDINGS

From our interviews, we found:

- Some documents take longer to obtain or correct and should be noted,
- Users may not always know how to resolve document errors,
- Applicants need AI feedback to understand the context of their application and have the ability to catch small errors, and
- Applicants distrusted the accuracy of AI validation **without human intervention**.

COMMON ERRORS

From the cure email LLM analysis, we found that **applicants made many low-severity and preventable errors**, such as:

- Missing information (ex: signature, name, date, etc.)
- Document discrepancy (ex: name mismatches, different points of contact)
- Document expiration

Knowing exactly what errors to focus on helped the team narrow the tool's scope.

PRODUCT DECISIONS

Or in other words: why does the product look this way / do this thing?

WHERE SHOULD VALIDATION HAPPEN?

1. Differing opinions on document validation location
   - At the beginning, point of upload, end (before submission), or standalone?
2. Joint brainstorming session in **Mural**
3. Mutually voted on a tool that **validates documents as they are uploaded throughout** the EDA application
4. Drafted a **product decision document**, noted cost/benefit analysis + pros/cons for each option thanks to team suggestions
   - Finalized recommendation, metrics, and milestones

HOW SHOULD WE MEASURE SUCCESS?

- Metrics considered: % of applicants who used the tool, % of errors caught upfront
    - Technical and operational constraints (API, stakeholder infrastructure)
- Instead, **work backwards from what we can control**
    - Built-in tool = track our own data without external dependencies
    - **# of docs flagged / # of docs submitted overall = % of errors**

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LESSONS LEARNED

- UX research → Product Fellow is a bigger step than I imagined
    - Managing more than just users, lots of alignment between multiple teams
- Team norming and facilitation play a big role in project success
        - Led a team norming session on **Mural** and facilitated weekly standups
- This is something I want to keep doing! <3

THANK YOU!

Shoutout to **Coding it Forward** (Cassie), **NJIA fellows** (Stephen, Will-Josh, Nate), and **my team** (Tristan, Curtis, Liz, Aleks, Zarak)!